ROSSMANN STORE SALES

Combined Analytics Report

Data Cleaning • Exploratory Data Analysis • Sales Prediction

Project	Rossmann Store Sales
Report date	12 November 2026
Data period	2013-01-01 to 2015-07-31
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PART 1 OF 3

Data Cleaning Report

Data quality assessment, treatment and validation of the Rossmann datasets

Project	Rossmann Store Sales
Stage	Data Cleaning
Datasets	Train, Test, Store, Sample Submission
Report date	12 November 2026

1,017,209	0	0	13
Training rows	Missing cells after cleaning	Duplicate rows	Store columns after cleaning

1. Dataset Overview

Dataset	Rows	Columns	Duplicate Rows	Missing Cells
Train	1,017,209	9	0	0
Test	41,088	8	0	11
Store	1,115	10	0	2,343
Sample Submission	41,088	2	0	0

2. Missing Values Analysis

Dataset	Column	Missing Count	Missing (%)
Test	Open	11	0.03
Store	CompetitionDistance	3	0.27
Store	CompetitionOpenSinceMonth	354	31.75
Store	CompetitionOpenSinceYear	354	31.75
Store	Promo2SinceWeek	544	48.79
Store	Promo2SinceYear	544	48.79
Store	PromoInterval	544	48.79

3. Duplicate Analysis

Dataset	Duplicate Rows Found
Train	0
Test	0
Store	0

4. Missing Value Handling

- **CompetitionDistance:** median imputation, with a missingness indicator created.
- **Promo2 fields:** missing values handled as structurally not applicable.
- **Competition opening information:** handled using indicators.
- **Test Open:** missing values inferred using Store and DayOfWeek history.

5. Outlier Analysis

Column	Q1	Q3	IQR	Lower Bound	Upper Bound	Outlier Count	Outlier (%)
Sales	4,859.00	8,360.00	3,501.00	-392.50	13,611.50	30,769	3.64
Customers	519.00	893.00	374.00	-42.00	1,454.00	40,853	4.84

- Outliers were detected using the IQR method.
- Genuine business outliers were retained.
- No automatic deletion or capping was performed.

6. Before and After Comparison

Dataset	Rows Before	Rows After	Cols Before	Cols After	Missing Before	Missing After	Dupes Before	Dupes After
Train	1,017,209	1,017,209	9	9	0	0	0	0
Test	41,088	41,088	8	8	11	0	0	0
Store	1,115	1,115	10	13	2,343	0	0	0

7. Final Data Quality Validation

Validation Check	Count	Status
Train missing cells	0	**PASS**
Test missing cells	0	**PASS**
Store missing cells	0	**PASS**
Train exact duplicates	0	**PASS**
Test exact duplicates	0	**PASS**
Store exact duplicates	0	**PASS**
Invalid train dates	0	**PASS**
Invalid test dates	0	**PASS**
Test Open missing	0	**PASS**
Negative Sales	0	**PASS**
Negative Customers	0	**PASS**
Invalid train DayOfWeek	0	**PASS**
Invalid test DayOfWeek	0	**PASS**

8. Cleaning Quality Summary

Dataset	Rows	Columns	Missing Values	Duplicate Rows
Train	1,017,209	9	0	0
Test	41,088	8	0	0
Store	1,115	13	0	0

9. Exported Cleaned Datasets

Datasets	`cleaned_train.csv`
	`cleaned_test.csv`
	`cleaned_store.csv`

10. Conclusion

The Rossmann datasets were cleaned and validated. Missing values, duplicates, data types, business rules and outliers were analysed. The cleaned datasets are ready for further Exploratory Data Analysis and Sales Prediction.

PART 2 OF 3

Exploratory Data Analysis Report

Sales patterns, store performance, promotions and customer behaviour

Project	Rossmann Store Sales
Stage	Exploratory Data Analysis
Period covered	2013-01-01 to 2015-07-31
Report date	12 November 2026

5.87 bn	5,773.82	+38.77%	0.824
Total sales	Average daily sales	Sales uplift with promotion	Customer-sales correlation

1. Dataset Overview

Metric	Value
Total records	1,017,209
Total stores	1,115
Total columns	25
Date range	2013-01-01 to 2015-07-31

2. Sales Summary

Metric	Value
Total sales	5,873,180,623.00
Average sales	5,773.82
Median sales	5,744.00
Maximum sales	41,551.00

3. Customer Summary

Metric	Value
Average customers	633.15
Maximum customers	7,388

4. Store Performance

Metric	Value
Total stores	1,115
Best performing store type	b
Highest sales month	3 (March)

5. Promotion Analysis

Metric	Value
Average sales without promotion	5,929.41
Average sales with promotion	8,228.28
Percentage difference	38.77%

6. Correlation Analysis

Variable Pair	Correlation
Customers vs Sales	0.824

7. Business Insights

- Promotion performance was analysed.
- Store type performance was compared.
- Monthly sales patterns were identified.
- Customer and sales relationships were examined.

8. Conclusion

The exploratory analysis identified sales patterns, store performance differences, promotional trends and time-based variations. These findings can support business decisions related to inventory planning, promotions and store performance improvement.

PART 3 OF 3

Sales Prediction Report

Machine learning models for daily store sales forecasting

Project	Rossmann Store Sales Prediction
Task	Task 3 - Sales Prediction
Selected model	Tuned Random Forest Regressor
Report date	12 November 2026

1,177.08	794.05	0.8516	41,088
RMSE (best model)	MAE (best model)	R^2 score (best model)	Test predictions generated

1. Project Overview

Objective: To predict daily store sales using Machine Learning models based on historical sales and store information.

2. Dataset Information

Item	Value
Training records	1,017,209
Testing records	41,088
Store records	1,115
Training features	27
Training date range	2013-01-01 to 2015-07-31

3. Data Preprocessing

- Cleaned datasets were used.
- Training and testing datasets were merged with store data.
- Missing values were checked.
- Categorical variables were encoded.
- Numerical features were retained.

4. Feature Engineering

- Year
- Month
- Day
- Day of Year
- Week of Year
- Weekend Indicator
- Month Start and Month End
- Competition Age
- Promotion Activity

5. Model Development

The following Machine Learning models were developed:

- Linear Regression
- Random Forest Regressor
- Tuned Random Forest Regressor

6. Model Evaluation

Evaluation metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and R^2 Score.

Model	RMSE	MAE	R^2
Tuned Random Forest	1,177.08	794.05	0.85
Random Forest	1,363.03	928.15	0.80
Linear Regression	2,664.04	1,946.24	0.24

7. Best Model

Selected Model	RMSE	MAE	R^2 Score
Tuned Random Forest	1,177.08	794.05	0.8516

8. Hyperparameter Tuning

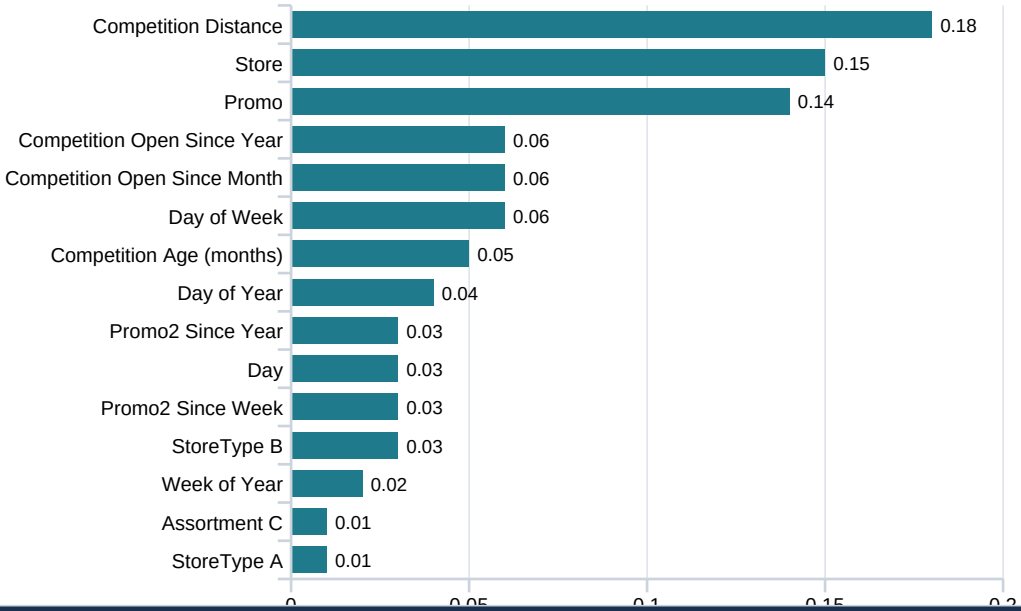
RandomizedSearchCV was used for Random Forest hyperparameter optimisation.

Best Parameter	Value
n_estimators	80
min_samples_split	5
min_samples_leaf	2
max_features	0.8
max_depth	25

Best cross-validation RMSE: 2,175.26

9. Feature Importance

Top 15 features ranked by importance:



Rank	Feature	Type	Importance
1	Competition Distance	Numerical	0.18
2	Store	Numerical	0.15
3	Promo	Numerical	0.14
4	Competition Open Since Year	Numerical	0.06
5	Competition Open Since Month	Numerical	0.06
6	Day of Week	Numerical	0.06
7	Competition Age (months)	Numerical	0.05
8	Day of Year	Numerical	0.04
9	Promo2 Since Year	Numerical	0.03
10	Day	Numerical	0.03
11	Promo2 Since Week	Numerical	0.03
12	StoreType B	Categorical	0.03
13	Week of Year	Numerical	0.02
14	Assortment C	Categorical	0.01
15	StoreType A	Categorical	0.01

10. Validation Predictions

Total validation predictions: 175,084. Actual and predicted sales were compared to evaluate model performance.

11. Final Test Predictions

Total test predictions: 41,088. Final sales predictions were generated for the test dataset.

12. Output Files

Models	`linear_regression_model.pkl`
	`random_forest_model.pkl`
	`tuned_random_forest_model.pkl`
	`best_sales_prediction_model.pkl`

Predictions	`model_comparison.csv`
	`validation_predictions.csv`
	`final_sales_predictions.csv`
	`feature_importance.csv`

Visualisations	`01_model_comparison.png`
	`02_actual_vs_predicted.png`
	`03_residual_distribution.png`
	`04_feature_importance.png`

Reports	`Sales_Prediction_Report.txt` `Sales_Prediction_Report.pdf`

13. Conclusion

Three Machine Learning models were developed and evaluated for Rossmann store sales prediction. The best-performing model was selected based on validation performance. Feature importance analysis was performed to identify important sales prediction factors, and final predictions were generated for the test dataset.